Accelerating Frontier MoE Training With 3-D Integrated Optics

Mikhail Bernadskiy, Peter Carson, Thomas Graham, Taylor Groves, and Ho John Lee, *Lightmatter, Mountain View, CA, 94040, USA*

As traditional semiconductor scaling slows, high-speed interconnects have emerged as the new scaling engine, enabling the creation of larger logical GPUs by linking many GPUs into a single, low-latency, high-bandwidth compute domain. While initial scale-up fabrics leveraged copper interconnects for their power and cost advantages, the maximum reach of passive electrical interconnects (approximately 1 m) effectively limits the scale-up domain to within a single rack. The advent of 3-D-stacked optics and logic offers a transformative, power-efficient scale-up solution for connecting hundreds of GPU packages across multiple data center racks. This work explores the design tradeoffs of scale-up technologies and demonstrates how frontier large language models necessitate novel photonic solutions to achieve aggressive power and performance targets. We model the benefits of 3-D integrated optics (Passage)-enabled GPUs and switches within the scale-up domain when training frontier mixture of experts models exceeding one trillion parameters. We achieve a 2.7× reduction in time-to-train.

The race to build larger, more sophisticated artificial intelligence (AI) models is pushing the limits of existing infrastructure. At the chip and package level, GPUs are constrained by shoreline, yields, and power. These challenges have led to the development of large high-bandwidth, low-latency scale-up pods. These pods effectively combine hundreds of GPUs into a single logical GPU to facilitate a variety of parallelism strategies (e.g., data, tensor, expert) for large AI models. Approaches like mixture of experts (MoE)[1] have pushed scale-up networks to their limits due to copper reach (1 m), which constrains the number of GPUs connected within a single network hop.

With MoE, an ensemble of specialized subnetworks work together through sparse activations to increase model capacity without significantly increasing computational requirements. The output of the selected experts are combined to create the final result; this relies on expensive all-to-all communications. Larger scale-up domains directly translate into the capability to deploy a larger number of experts and improve model performance.

This article explores the limitations of current approaches and presents a paradigm shift: the transition from copper to integrated 3-D photonics in order to create a more scalable and efficient high-bandwidth domain across racks in the data center. In this work, we show the following:

- Passage 3-D photonics technology has the unique combination of bandwidth density, port count, reach, and energy efficiency to enable multiple generations of innovation in AI (8× scale-up bandwidth, half the energy).
- Comparisons of system design tradeoffs using electrical, linear pluggable optics (LPO), copackaged optics (CPO), and 3-D integrated optics (3DCPO) (7× reduction in package area).
- Application benefit of an expanded scale-up domain for large language model (LLM) training (2.7× speedup).

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BACKGROUND

LLM Training

Transformer-based language models have steadily increased in size, with higher parameter counts enabling increasingly powerful model capabilities. Recent frontier models have on the order of 1 trillion parameters and are trained on datacenter-scale clusters. The compute and memory requirements for training a transformer are dominated by the attention block and the feed-forward network (FFN) in each layer, which are mostly matrix multiplication operations.

In sparse MoE models,[1] the FFN layer in the original (now referred to as "dense") transformer is replaced by multiple experts, and a small additional routing network selects which and how many alternatives should be activated for each token. This enables a larger, potentially more expressive model size at a given amount of compute with an additional pattern for routing tokens to selected experts within an MoE layer (expert parallelism).

Scale-Up Networking

Historically, the GPU interconnect bandwidth was limited by Peripheral Component Interconnect Express (PCIe). The advent of NVLink 1.0 for the Pascal generation of GPUs allowed for a limited number of GPUs to create a high-bandwidth scale-up domain at five times the PCIe bandwidth. This was a massive leap forward in bandwidth and effectively enabled the multi-GPU tensor parallelism prominent in modern training.

As the number of GPUs increased, switches were incorporated into the design to facilitate the increased bandwidth between a larger number of accelerators. Scale-up topologies generally follow one of two approaches. Either a multidimensional torus or a single layer of switching (SLS), which uses multiple GPU rails to switch connections. The SLS is inherently a low-latency network with deterministic routing and performance. This allows full bandwidth between any pair of GPUs, but the scale of the network is limited to the number of ports on the switch (i.e., a 512-port switch can support at most 512 GPUs—one port per GPU). Because of these characteristics, we focus on the SLS topology for the purposes of this article.

The size of the GPU scale-up domain has continued to increase over time. One hundred forty-four radix scale-up switches have been announced to support 144 GPU packages in 2027. The limiting factor in scaling the pod beyond 144 packages has been the reliance on copper and electrical networking. While electrical networking provides benefits in terms of simplicity and energy efficiency, the reach limitations of high data rate SerDes means an electrically connected GPU pod is effectively limited to one or two racks. As the size and bandwidth demands of the scale-up network continue to increase over time, 3DCPO provide the ultimate solution, with the bandwidth density and energy efficiency of electrical networks but longer reach.

Motivation for 3DCPO

Package Growth and Shoreline Limitations

Slowdowns to Moore's Law and Dennard Scaling have necessitated larger packages and increased power to deliver next-generation GPU performance. As packages grow, computational capability grows proportional to the area while the input–output (I/O) is limited to the perimeter of the GPU. To complicate matters, large portions of the shoreline are reserved for High Bandwidth Memory (HBM), which require short trace lengths for signal integrity.

Figure 1 shows an example of a GPU package where four logic chips, 16 stacks of HBM, and I/O dies are all placed on a substrate. Both I/O and HBM compete for the shoreline of the chip, leaving only the east and west available for scale-up bandwidth. For each I/O die, the bandwidth is limited by the number of SerDes macros that can fit along an edge. Doubling the bandwidth of these SerDes from 224 Gb/s to 448 Gb/s creates signal integrity challenges which require sophisticated equalization and increased power.

Electrical Reach Limitations

As speeds of SerDes increase, electrical reach of the signal is reduced. At 224 Gb/s, the reach of passive direct attached copper (DAC) is approximately 1 m, and at 448 Gb/s the reach is expected to be tens of centimeters. To reduce the insertion loss, high-speed electrical solutions are moving toward copackaged copper (CPC), and flyover cables to bypass lossy printed circuit board (PCB) traces. For longer distances, retimers must be deployed, which increase power, area, and cost. The short reach of copper means GPUs and switches must be densely configured within a single rack. This creates rack-level power challenges and shifts costs to cooling and infrastructure.

Challenges for Existing Optical Solutions

The hurdles to deploying optics broadly within the datacenter have been cost, reliability, and energy efficiency. While optics cost more than passive copper solutions, the benefits of expanded reach add the potential for increased GPUs in the scale-up domain and

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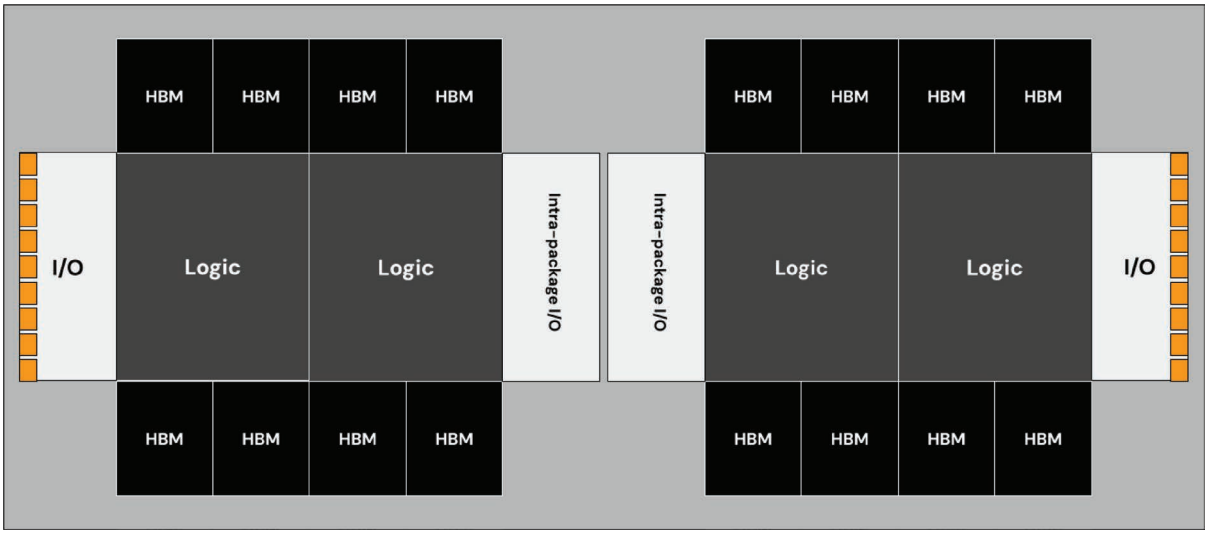


FIGURE 1. A GPU package in a 4 × 1 reticle configuration. Four logic reticles surrounded by HBM stacks on the north and south side in black, intra-package I/O in the middle and inter-package I/O on the east and west side. SerDes shoreline is highlighted in orange.

faster time to solution. We demonstrate this benefit in the "Results" section. Optics typically use lasers to power the transmission. Lasers add cost, power, and can be temperature sensitive, failing at higher rates compared to copper connections. Laser solutions must have fault tolerance and field-replaceable features baked into the design when operating at datacenter scale.

Optics enable disaggregation of the scale-up pod, creating an opportunity for power and cooling savings at the rack level, but given the massive amount of scale-up bandwidth (order of magnitude greater than scale-out), optics must be incredibly power efficient to fit within the GPU package and tray power budgets. At 5 pJ/bit, optics is effectively at parity with passive copper based solutions[2] and 14.4 Tb/s of scale-up bandwidth results in 72 W of power per GPU. At 20 pJ/bit, this increases to more than 288 W per GPU and reduces power available to computation. Twenty pJ/bit effectively makes higher levels of scale-up bandwidth infeasible. Energy efficiency of the scale-up network is paramount.

Optical modules: Typical pluggable optical modules (e.g., OSFP) often integrate power-hungry digital signal processor (DSPs) and retimers to overcome host-to-module signal loss, resulting in high aggregate power (e.g., 21 pJ/bit) and large form factors (> 2000). While easily field-replaceable and interoperable across platforms, their inherent power consumption and significant area footprint limit density.

LPO: LPO transceivers are an optimization of conventional pluggable optics modules such that the DSP is removed from the module itself. It is a linear drive in the sense that the signal from one host to another host device does not require a retimer or incur the extra power and performance overheads. The expectation is that an LPO module is approximately 25%–50% more power efficient than a conventional pluggable module. This creates a reliance on the host-side interface to do the heavy lifting and drive the signal without retimers. Therefore host SerDes in an LPO-based system are expected to rely on DSPs and be in the range of 4.5–6 pJ/bit.[2] LPO solutions must be co-designed in consideration of the host platform capabilities and link budget specific to a given end device (GPU, CPU, switch, and so on). In some cases, LPO utilizes flyover cables between the host and the module to reduce losses further, but this adds expense and complexity. LPO still leverages large form factors (e.g., OSFP-XD) resulting in low bandwidth density compared to integrated optics. As data rates and the number of channels per module increase the modules may require cold-plate cooling.

CPO: CPO describes the process of taking the optical transceiver and moving it onto the same package as the device it is supporting (typically a processor or switch). It is compelling because you reduce the distance traveled electrically over a high loss medium such as a PCB, and decrease latency compared to a pluggable optical module. When discussing CPO, it is helpful

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to distinguish between the approach taken to build the optical engine (OE) and how it's integrated into the host package. In both contexts, the concepts of 2-D, 2.5-D, and 3-D design can apply. A good overview of this topic has been provided by Lee, et. al.[3] In both 2-D and 2.5-D packaging, the host chip and OE are placed side by side, but 2-D integration uses an organic substrate with further distance between the OE and host whereas 2.5-D has higher bandwidth density and energy efficiency. Two-dimensional approaches can result in larger packages and greater beachfront expansion as traces fan-out from the host to OEs. The losses associated with the beachfront expansion translate into increased energy consumption on the host SerDes.

PASSAGE

In contrast to the previously mentioned solutions, Lightmatter Passage is a 3-D photonics platform where optical elements are located directly underneath the footprint of electrical SerDes. This creates a tightly integrated, high-bandwidth, and low energy solution. The energy efficiency of current Passage products is 2.3 pJ/bit for photonic integrated circuit (PIC) and laser plus SerDes. This is significantly more efficient as the electrical signal only travels a short distance vertically. For short reach SerDes [e.g., extreme short reach (XSR) or very short reach (VSR)] this may be 1 pJ/bit[4] at 112 Gb/s PAM-4 or 2 pJ/bit with non-return-to-zero (NRZ) modulation. This results in substantially greater efficiency than competing solutions when considering the entire package [(PIC), electronic integrated circuit (EIC), laser, and host SerDes].

Passage is offered as either 1) 3-D chiplet/OE where the PIC sits underneath an I/O die or 2) an optical interposer that sits under the entirety of the processor or switch. Figure 2 shows an exploded view of a Passage interposer design, where the EIC ① sits on top of the PIC ②. The EIC consists of one or more reticles and can be as large as a waferscale. While a traditional EIC places I/O and SerDes along its perimeter, 3-D stacking allows the EIC to utilize I/O from anywhere within the chip area as indicated by vertical blue bars (signals) in the image. The PIC does the electrical-to-optical conversion and modulates the signal using microrings that sit 100 microns beneath the SerDes. Up to 16 signals are muxed across different wavelengths and travel through a circuit-switched, waveguide network to another SerDes on the chip or externally via detachable optical fibers. Optical power is provided from an external laser. The PIC contains through-silicon vias to provide the EIC with power and signaling from the substrate ⑤. For a more detailed description see Bunandar.[5]

SYSTEM DESIGN WITH PASSAGE

In this section, we examine three different approaches to constructing an optical GPU solution that enable a 512-GPU package (2048-GPU die) scale-up Pod. The approaches are 1) LPO, 2) 2.5-D CPO with 2-D integration, and 3) Passage optical interposer. For each approach, we generate projections of the power and energy required as well as the growth in area (package and board). We assume a single-layer-of-switches (SLS) topology as explained in the "Background" section, such that each switch has at least one port connected to every GPU in the pod.

Port Definition

We assume 448-Gb/s raw bandwidth per port. Larger port designs make it challenging to build high radix switches, as the aggregate bandwidth within the switch fabric increases. Smaller port designs lead to inefficient use of data fibers and poor bandwidth density. A 400-Gb/s port can be constructed differently dependent on the SerDes speed and number of lanes per port. Lightmatter has demonstrated 16 λ bidirectional at 56-Gb/s NRZ encoding on a single fiber. Being able to interleave transmit and receive on a single fiber doubles the port count and fiber density possible for a given shoreline. Alternatively, the link could consist of four lanes of 112 Gb/s, or two lanes of 224 Gb/s.

Energy Efficiency

The energy efficiency values of this section are identical to those referenced in the work of Bernadskiy et al.[2] as follows:

- 224G PAM-4 LR SerDes (host-package): 5 pJ/b
- 56G NRZ XSR SerDes (host-package): 2 pJ/b
- 1.6T DR8 LPO (off-package): 8 pJ/bit
- 2.5D CPO (host-package): 4.7 pJ/b
- ELSFP laser (off-package): 2.3 pJ/b
- Passage PIC (host-package): 1.2 pJ/b
- Passage laser (off-package) 1.1 pJ/b.

There are substantial power savings available by 3-D stacking the PIC and EIC allowing as Passage does, which allow for the shortest electrical channel and reduces both SerDes electrical and optical power requirements. Other approaches require longer channel lengths with greater losses, which lead to power-hungry equalization and DSPs to compensate. Figure 3 shows the aggregate in-package and out-of-package power for a 32-Tbps GPU for the three optical solutions, with Passage demonstrating more than 200 W in savings.

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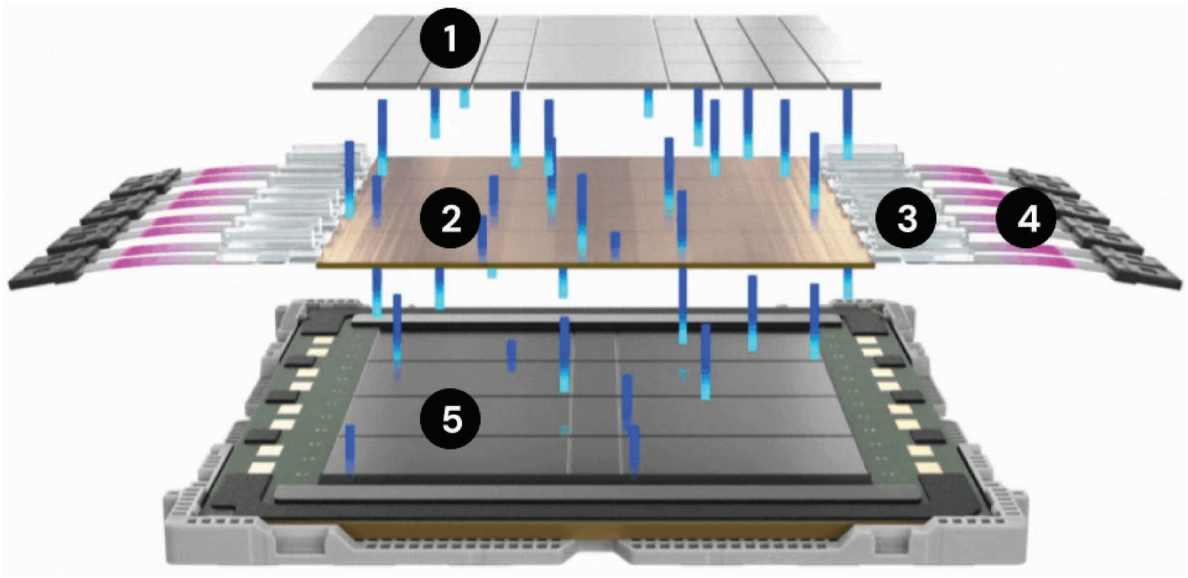


FIGURE 2. Exploded view of Passage 3-D optical solution: ① EIC, ② Passage PIC, ③ fiber attach unit, ④ fibers, and ⑤ substrate. Blue vertical lines distributed throughout package area represent I/O enabled without SerDes shoreline constraints.

Area Estimates

LPO

For the LPO modules, we use the specified 105.8 mm × 22.58 mm dimensions[2] for a total area of 2389 mm^2 per module. We assume up to 16 channels (32 fibers) within a single extra dense module. For a 3.2 T module, this results in an areal bandwidth density of 1.3 Gb/s/mm^2.

2.5-D, 2-D Integrated CPO

For this analysis we assume a 15 mm × 25 mm footprint for an OE with 10 mm of beachfront and 12.8-Tbps unidirectional bandwidth (using 224 G SerDes).[2] This suggests areal density of approximately 34 Gb/s/mm^2 or 24 Gb/s/mm^2 when accounting for beachfront.

Passage Interposer

When examining a design with an optical interposer, the primary source of package growth is area for attaching optical fibers to. The fiber attach area expands the PIC 5 mm beyond the host chip. The other dependency is the number of fibers being attached. These fibers are 127 μm and can be estimated at four data fibers per millimeter of shoreline for a single array of fibers. After accounting for laser fibers, this translates to roughly 1350 Gbps/mm and 270 Gb/s/mm^2 for a 16 λ 56 G bidirectional link that interleaves 8-λ TX and 8-λ RX on a single data fiber.

Impact on GPU Design

GPU

For a GPU configuration similar to Figure 1 with 16 stacks of HBM4 (north and south sides) totaling 209 Tb/s (26 TB/s) of memory bandwidth (6.4 GT/s). We assume a full reticle is 26 mm × 33 mm and that stacks of HBM are 13 × 11 mm. Given, eight stacks of HBM on each side of the GPU in Figure 1, we assume an 88-mm (8×11) package in the *x*-dimension for simplicity. This leaves two sides of the package available for I/O. For I/O we assume 32 Tb/s TX and 32 Tb/s RX

Figure 3 shows three different approaches to providing 32 Tb/s of optical bandwidth on a GPU. This would require 160 channels of 224-Gb/s-PAM4. This is equivalent to 10 OSFP-XD modules. In aggregate, this is more than 23,000 mm^2 of board area. Additionally, the bandwidth required per device would likely lead to the use of CPC and flyover cables from the host to the modules to reduce PCB losses. CPC area is approximately 2 × 2 mm per copper differential pair. This means that a 4×4 CPC connector is 8 mm × 8 mm and there would be five on each side of the GPU. This results in 18% growth in the *x*-dimension of the GPU.

For a 2.5-D CPO solution using the areal bandwidth densities previously calculated, this would result in approximately 35 mm of package expansion off each end of the GPU, or 80% package expansion in the *x*-dimension in aggregate.

For a Passage design with bidirectional links, this is a relatively small 118 mm^2 worth of fiber attach or 5%

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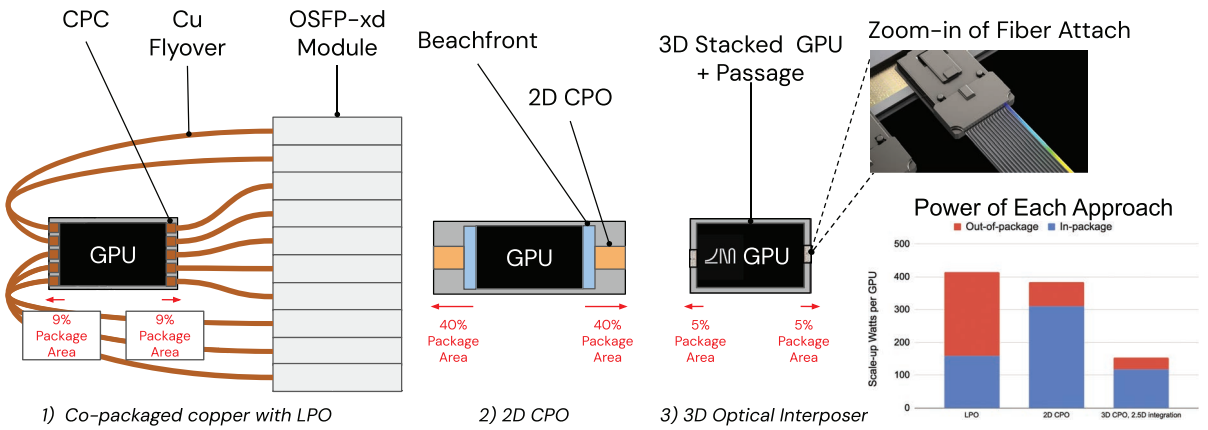


FIGURE 3. Illustration showing the approximate area and power required for a 32-Tbps GPU with 1) LPO with CPC flyover cables, 2) 2-D integrated CPO with beachfront expansion, and 3) 3-D stacked GPU with Passage Optics.

substrate growth in the *x*-dimension on each side of the GPU (10% if attaching to both sides of the GPU) This is seven times less package expansion in the *x*-dimension than conventional CPO.

APPLICATION MODELING

Analytical Modeling Tool

We developed `ScaleScope`,[a] an analytical performance modeling tool for LLMs that enables rapid evaluation of different architectures and deployment strategies. Following established approaches for LLM and MoE performance modeling,[6] the tool models execution time as a combination of computation, memory access, and communication costs. `ScaleScope` decomposes LLM execution into constituent operations—attention, feed-forward networks, and expert routing—while accounting for hardware capabilities, system topology, and parallelization strategies including data parallelism (DP), tensor parallelism (TP), pipeline parallelism (PP), and expert parallelism (EP). Communication operations are modeled using the Hockney model,[7] capturing both fixed overhead and bandwidth-dependent transfer costs for collective operations on various topologies.

Parallelism Strategy

To handle the architectural heterogeneity between dense attention layers and sparse MoE layers, we employ a decoupled parallelism strategy that applies different mappings to each while cohosting them on the same GPUs.

[a]https://github.com/lightmatter-ai/scalescope

Attention layers use a `TP` × `DP` × `PP` mapping: TP splits attention computations across devices within each layer, DP replicates weights across groups processing different batch samples, and PP assigns different layers to different device groups for pipelined execution.

MoE layers use a `ETP` × `EP` × `EDP` × `PP` mapping based on the distribution of the full expert set (FES)—a complete collection of all experts. A single FES is hosted across `ETP` × `EP` devices, where EP partitions experts (one per EP unit) and expert tensor parallelism (ETP) shards each expert's computation across multiple devices. This FES is then replicated `EDP` times for data parallelism, while PP is shared with the attention mapping.

Figure 4(a) illustrates this cohosting: The attention layer is tensor-parallelized across all four GPUs (TP = 4), while experts are distributed using ETP = 2, with Expert 0 on GPU0–1 and Expert 1 on GPU2–3.

Figure 4(b) shows the complete architecture across four DP ranks, with two replicated FESs (EDP = 2), each distributed across EP = 4 units. Three communication patterns emerge: EP for token routing between attention and experts, DP for attention gradient synchronization, and EDP for expert gradient synchronization across FES replicas.

Duplicate token dropping[8] eliminates redundant communication by having each TP rank retain only a unique subset of tokens before EP all-to-all exchanges, reducing communication volume by a factor equal to the TP degree—a critical optimization since EP communication often traverses slower internode links, which our modeling incorporates.

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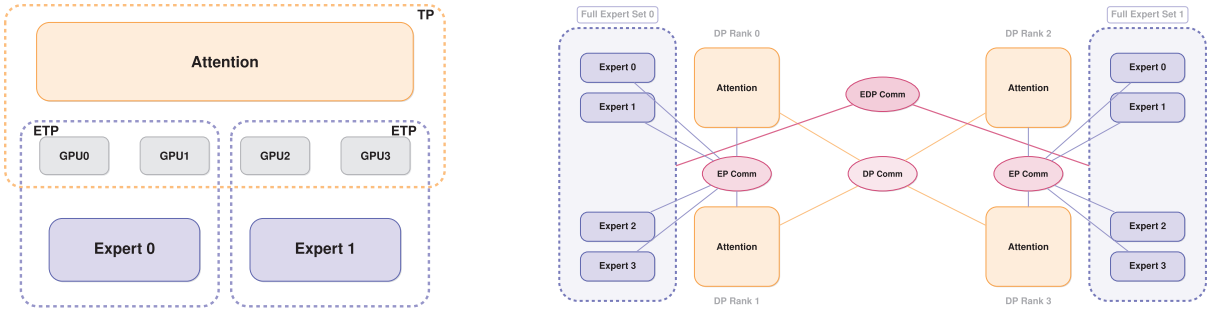


FIGURE 4. Parallelism strategies in distributed MoE. (a) TP and ETP layout for attention and expert layers. (b) Multidimensional parallelism and communication patterns.

Scaling MoE Architectures: Expert Count and Fine-Grained Segmentation

To make larger expert pools computationally feasible, fine-grained expert segmentation[9,10] has emerged as a crucial technique. This approach partitions each expert's feed-forward layer along its hidden dimension. Specifically, if an original expert operates on an intermediate dimension $d_{ff} = r{\cdot}d$ (typically with $r = 4$), fine-grained segmentation creates multiple smaller experts, each operating on dimension d_{ff}/s, where s is the segmentation factor.

By creating s times more experts (each s times smaller) and activating s times more experts per token, we maintain constant computational costs while enabling the benefits of larger expert pools. This transformation effectively enables sophisticated MoE architectures that would otherwise be computationally prohibitive.

Formally, we apply a segmentation factor s that transforms the original MoE configuration as follows:

- *Number of experts*: $N'_e = s \cdot N_e$ (more experts).
- *Activated experts per token*: $k' = s \cdot k$ (more activated experts).
- *FFN expansion factor*: $r = r/s$ (smaller per-expert hidden dimension).
- *Expert tensor parallel degree*: $T'_p = T_p/s$ (less parallelism needed per expert).
- *Number of accelerators per FES*: $N'_{\text{GPU}} = N_{\text{GPU}}$. (The number of accelerators per FES stays the same.)

We refer to this set of transformations as the *scaling transformations*.

This segmentation preserves the total compute while dramatically increasing the effective expert count, enabling exploration of MoE architectures with hundreds of experts.

Scaling Invariances

We demonstrate that coscaling architectural and parallelism parameters with segmentation factor s preserves not only total FLOPs[9,10] and parameter count,[10] but also per-accelerator invariances in computational load, activation memory, and intra-expert communication volume—properties essential for interpreting our simulation results. These invariances do not extend to EP communication, which scales with the number of experts.[9] For simplicity, we assume even token distribution among experts; load imbalance can be accommodated by introducing an imbalance factor.

We compare a *baseline* configuration with N_e experts, FFN expansion factor r, k active experts per token, and ETP degree T_p, against a *scaled* configuration with segmentation transformations from the "Scaling MoE Architectures: Expert Count and Fine-Grained Segmentation" section applied.

Computational Invariance

Token Load Distribution. With T tokens processed per FES, the expected tokens per expert equals total routed tokens divided by expert count as follows:

$$\text{Baseline: } \frac{T{\cdot}k}{N_e} \rightarrow \text{Scaled: } \frac{T{\cdot}(sk)}{sN_e} = \frac{Tk}{N_e}.$$

The segmentation factors cancel, proving token load per expert is invariant.

FLOPs per expert: Each expert implements a two-layer FFN costing $4d \cdot d_{ff}$ per token routed to it. Since token load is invariant, scaling to $d'_{ff} = d_{ff}/s$ reduces per-expert FLOPs by a factor of s.

Total FES FLOPs: The baseline total is $N_e \times (Tk/N_e \cdot 4d \cdot d_{ff}) = 4Tkrd^2$. The scaled total becomes $(sN_e) \times (1/s)\,(Tk/N_e \cdot 4rd^2) = 4Tkrd^2$. The factors s and $1/s$ cancel – total FLOPs are invariant, and with fixed N_{GPU}, per-GPU computational load is preserved.

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Memory Invariance

Parameter memory: Each expert has $2(d \cdot d_{ff}) = 2rd^2$ parameters. Total FES parameters: baseline $N_e \times 2rd^2$, scaled $(sN_e) \times 2(r/s)d^2 = 2N_e rd^2$ – invariant. Per-GPU parameter memory is therefore the same.

Optimizer states and gradients: These scale linearly with parameters, hence are also invariant on both total and per-GPU basis.

Input activations: With ETP, each GPU hosts an expert shard but must receive all tokens destined for that expert before processing. This remains Tk/N_e tokens of dimension d in both configurations.

Intermediate activations: Megatron-style TP stores activations after the first projection, sharded across ETP devices as follows:

$$\text{Baseline: } \frac{Tk \cdot d_{ff}}{N_e \cdot T_p} \rightarrow \text{Scaled: } \frac{T(sk) \cdot (d_{ff}/s)}{sN_e \cdot (T_p/s)} = \frac{Tk \cdot d_{ff}}{N_e \cdot T_p}.$$

The segmentation factors cancel, preserving per-GPU memory.

Memory bandwidth: Since per-GPU memory for parameters, optimizer states, gradients, and activations are all invariant, memory bandwidth requirements per GPU are preserved.

Communication Invariance

ETP communication: All-reduce volume within an expert scales with activation volume times $T_p - 1/T_p$. For the scaled configuration, this becomes $(T_p/s - 1)/(T_p/s) = T_p - s/T_p$, which decreases with s but approaches invariance for larger T_p.

Expert data parallelism (EDP) communication: EDP involves all-reduce for gradient synchronization across replicas. Since per-accelerator parameter count is invariant, EDP communication volume is unchanged.

EP Communication Scaling

EP requires all-to-all communication for token routing, with volume proportional to $T{\cdot}k{\cdot}d$.[9] With $k = sk$ in the scaled configuration, this becomes $T \cdot (sk) \cdot d$, scaling linearly with s. EP communication is therefore *not* invariant under our scaling strategy.

RESULTS

We evaluate MoE configurations in a large-scale training setup: a 120-layer decoder-only transformer with $d = 12288$, 128 attention heads, and 4.7 T total parameters, following the GPT architecture family. The model uses Megatron-style tensor parallelism[11] and mixed-precision training (BF16 compute/gradients, FP32 master weights and Adam optimizer states).

Training runs on 32,768 GPUs (8.5 PFlops BF16 each, 1600 Gb/s unidirectional scale-out Ethernet) with fixed parallelization: TP = 16, DP = 256, PP = 8, FES = 512, EDP = 8. We use global batch size 4096, sequence length 8192, targeting 13 T training tokens.

We evaluate these configurations across two distinct pod sizes of 128 and 512 GPU packages as follows:

- A network with 256 scale-up pods, each with 128 active GPUs packages and 14.4 Tb/s unidirectional bandwidth per GPU.
- Passage: An optical network with 64 scale-up pods, each with 512 active GPU packages and 32 Tb/s unidirectional bandwidth per GPU.

Within these fixed infrastructure constraints, we explore different MoE scaling strategies across four configurations, referred to as Config 1–4. The expert segmentation factor $s \in \{1, 2, 4, 8\}$ determines how each configuration implements fine-grained experts: Config 1 ($s = 1$) uses standard experts with full d_{ff} hidden dimension, while each subsequent configuration splits experts into progressively smaller units—Config 4 ($s = 8$) divides each original expert into eight fine-grained experts with hidden dimension $d_{ff}/8$. As s increases, active/total experts scale proportionally (1/32, 2/64, 4/128, 8/256) while ETP size decreases inversely (16, 8, 4, 2), maintaining constant computational workload.

The performance comparison between Passage and the alternative solution reveals significant differences in how these architectures scale across different MoE configurations. Our analysis focuses on relative performance scaling, using Config 1 of Passage as the baseline reference point.

To isolate the impact of network bandwidth differences between architectures, we first compare both systems assuming identical pod sizes of 512 GPU packages across 64 pods (32,768 GPUs total), as shown in Figure 5(a). As configurations progress, both systems experience the same dynamics: EP increases linearly with segmentation factor s while ETP decreases from T_p to T_p/s. As established in the "Scaling Invariances" section, this scaling approach preserves per-device computation, memory requirements, and EDP communication across all configurations. ETP communication decreases with s, with larger segmentation factors yielding more noticeable reductions.

Despite identical workloads, the bandwidth difference (32 Tb/s for Passage versus 14.4 Tb/s for the alternative) leads to divergent outcomes. Passage's high bandwidth allows communication/computation overlap. The alternative system cannot hide

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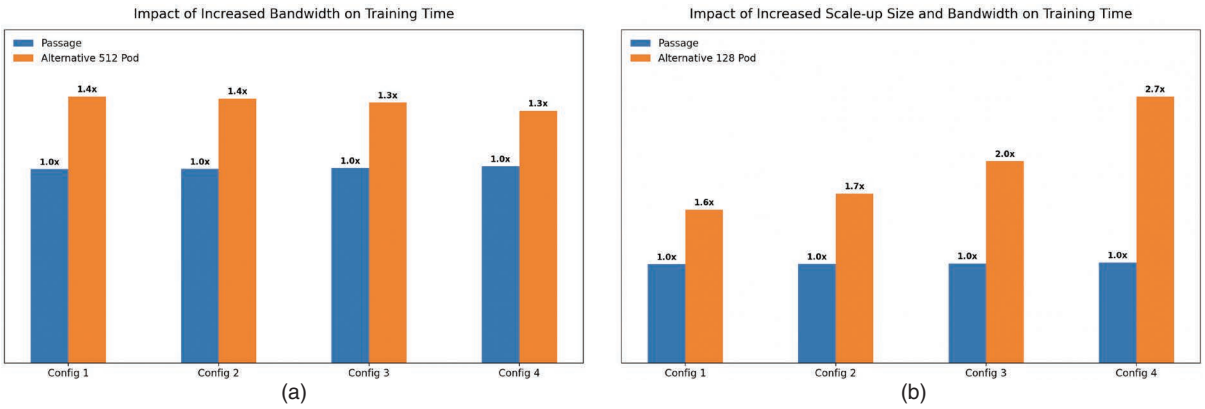


FIGURE 5. Training performance comparison between Passage and Alternative architectures under different radix configurations. (a) Relative performance of both architectures assuming the same radix-512 (normalized to Config 1 Passage baseline). (b) Relative performance with system-specific radix settings: Passage (512) versus Alternative (128) (normalized to Config 1 Passage baseline).

communication as effectively: Configs 1–2 require 1.4× longer training time, while Configs 3–4 improve to 1.3×. This improvement reflects the ETP behavior described earlier—reduced ETP costs more than compensate for increased EP communication.

These results demonstrate that while EP–ETP tradeoffs maintain invariant workloads, actual performance depends critically on network bandwidth. Passage's 32-Tb/s bandwidth achieves consistent training times across all configurations, whereas the alternative remains communication-bound at 14.4 Tb/s. Our modeling shows communication comprising 46%–51% of nonoverlapped execution time for the alternative versus 30%–34% for Passage, confirming communication is no longer the dominant factor in the latter.

When comparing systems at equivalent scale (32,768 GPUs total) but with architecture-specific network configurations—Passage using 64 pods of 512 GPUs at 32-Tb/s unidirectional versus the alternative using 256 pods of 128 GPUs at 14.4-Tb/s unidirectional—the performance gap widens [Figure 5(b)], particularly with finer-grained expert configurations, where experts per token increase (from 1 in Config 1 to 8 in Config 4). The alternative system requires 1.6× longer training time than Passage for Config 1, deteriorating to 2.7× for Config 4, while Passage maintains efficient scaling.

This scaling challenge stems from the expert all-to-all communication pattern, where tokens must be routed to their designated experts across the distributed system.[12] EP increases linearly with segmentation, so does the communication volume for token routing. The increased bandwidth and scale of Passage keeps EP at manageable levels. For Config 4, Passage spends 10.2% of total time in EP communication, while the alternative spends 60.3% of time in EP communication.

This efficiency has broader implications: Traditional MoE systems require careful tuning of load balancing and routing restrictions to prevent congestion. Passage's high-bandwidth architecture eliminates such constraints while maintaining stable performance, simplifying training and enabling more flexible expert utilization.

CONCLUSION

Our modeling demonstrates the profound impact of 3DCPO on the efficiency of MoE training. The results show that expanded radix and higher aggregate bandwidth of the 3-D optical interconnect deliver substantial performance gains. For scale-up pods of the same size, the higher bandwidth alone delivers up to 1.4× speedup. The performance gap widens further when comparing actual system configurations—Passage's 512-radix network versus the alternative's 128-radix topology. Here, Passage achieves a 2.7× speedup for the most demanding configuration (Config 4) by accommodating more EP communications within the high-bandwidth domain. Critically, Passage's elimination of communication bottlenecks ensures that additional compute capacity can be fully utilized rather than sitting idle waiting for data transfers—enabling higher computational intensity that would be wasted in bandwidth- and radix-constrained architectures. These findings underscore that MoE workloads effectively leverage Passage's expanded optical interconnect radix, accelerating traffic that would otherwise

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traverse slower scale-out networks. The combined benefits of higher bandwidth and connectivity enable Passage to maintain strong scaling efficiency even as expert counts and routing complexity increase. Future work will further optimize 3DCPO technology to leverage the full potential of high-radix optical interconnects and optical circuit switching.

REFERENCES

1. N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," 2017, *arXiv:1701.06538*.
2. M. Bernadskiy, P. Carson, T. Graham, T. Groves, H. J. Lee, and E. Yeh, "Accelerating frontier MoE training with 3D integrated optics," in *Proc. IEEE Symp. High-Perform. Interconnects (HOTI)*, 2025, pp. 25–35, doi: 10.1109/HOTI66940.2025.00021.
3. B. G. Lee, N. Nedovic, T. H. Greer, and C. T. Gray, "Beyond CPO: A motivation and approach for bringing optics onto the silicon interposer," *J. Lightwave Technol.*, vol. 41, no. 4, pp. 1152–1162, 2023, doi: 10.1109/JLT.2022.3219379.
4. D. Tonietto, "Pushing energy efficiency limits in wireline communication," presented at the SSCS Wireline Workshop, Toronto, Canada, 2022.
5. D. Bunandar, "Passage M1000: 3D photonic interposer for AI," 2025. [Online]. Available: https://hc2025.hotchips.org/assets/program/conference/day2/57-lightmatter_hotchips_2025_final.pdf
6. J. He et al., "FasterMoE: Modeling and optimizing training of largescale dynamic pre-trained models," in *Proc. 27th ACM SIGPLAN Symp. Princ. Pract. Parallel Program. (PPoPP)*, New York, NY, USA: ACM, 2022, pp. 120–134, doi: 10.1145/3503221.3508418.
7. R. W. Hockney, "The communication challenge for MPP: Intel Paragon and Meiko CS-2," *Parallel Comput.*, vol. 20, no. 3, pp. 389–398, 1994, doi: 10.1016/S0167-8191(06)80021-9. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167819106800219
8. S. Singh et al., "A hybrid tensor-expert-data parallelism approach to optimize mixture-of-experts training," in *Proc. 37th ACM Int. Conf. Supercomput. (ICS)*, New York, NY, USA: ACM, 2023, pp. 203–214, doi: 10.1145/3577193.3593704.
9. J. Krajewski et al., "Scaling laws for fine-grained mixture of experts," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 33,270–33,288.
10. D. Dai et al., "DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models," 2024, *arXiv:2401.06066*.
11. M. Shoeybi, M. Patwary, R. Puri, P. LeGresley, J. Casper, B. Catanzaro, "Megatron-LM: Training multi-billion parameter language models using model parallelism," 2020, *arXiv:1909.08053*.
12. D. Lepikhin et al., "GShard: Scaling giant models with conditional computation and automatic sharding," 2020, *arXiv:2006.16668*.

MIKHAIL BERNADSKIY is a senior staff AI software engineer at Lightmatter, Mountain View, CA, 94040, USA. His research interests include artificial intelligence (AI) workload modeling and photonic interconnects for AI training and inference. Bernadskiy received his Ph.D. degree in computer science from the University of Pennsylvania. Contact him at mikhail@lightmatter.co.

PETER CARSON is a senior director of Product Marketing at Lightmatter, Mountain View, CA, 94040, USA. His research interests include silicon photonics-based interconnects and the impact of network performance on communication-intensive artificial intelligence workloads. Carson received his B.S. degree in marketing from Manhattan University. Contact him at pcarson@lightmatter.co.

THOMAS GRAHAM is cofounder and head of Machine Learning at Lightmatter, Mountain View, CA, 94040, USA. His research interests include distributed systems and infrastructure for machine learning. Graham received his M.S. degree in computer science from Columbia University. Contact him at thomas@lightmatter.co.

TAYLOR GROVES is a principal solutions architect at Lightmatter, Mountain View, CA, 94040, USA. His research interests include high-performance computing, distributed systems, performance modeling, simulation, and interconnects. Groves received his Ph.D. degree in computer science from the University of New Mexico. He is a Member of IEEE. Contact him at taylor.groves@lightmatter.co.

HO JOHN LEE is a machine learning software architect at Lightmatter, Mountain View, CA, 94040, USA. His research interests include machine learning systems, photonic computing, and high-performance artificial intelligence software architecture. Lee received his M.S. degree in electrical engineering and computer science from the Massachusetts Institute of Technology. He is a Member of IEEE. Contact him at hojohn@lightmatter.co.

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